

복사전달 특강

Special Topics in Radiative Transfer

Lecture 5

2026 March 31 (Tuesday), 1PM

updated on 03/26

선광일 (Kwangil Seon)

KASI / UST

[Zeeman Effect]

- **Zeeman effect**

the splitting of atomic energy levels in the presence of an external magnetic field.

- **Zeeman effect at 1420MHz (21.1 cm)**

(1) The upper level of the 1420 MHz hyperfine transition of hydrogen is the $F = 1$ state. Its angular momentum has three possible projections onto a given axis. The three states are

$$F_z = m_F \hbar \quad (m_F = -1, 0, +1)$$

The three states have the same energy (i.e., they are degenerate).

However, if the atoms are immersed in an external magnetic field, the three states take on different energies. This is a consequence of the interaction of the magnetic moment of the hydrogen atom with the external field.

$$\begin{aligned} E_{\text{pot}} &= -\boldsymbol{\mu} \cdot \mathbf{B} \simeq g_e \frac{e}{2m_e} \mathbf{F} \cdot \mathbf{B} \\ &= g_e \frac{e\hbar}{2m_e} m_F B \end{aligned}$$

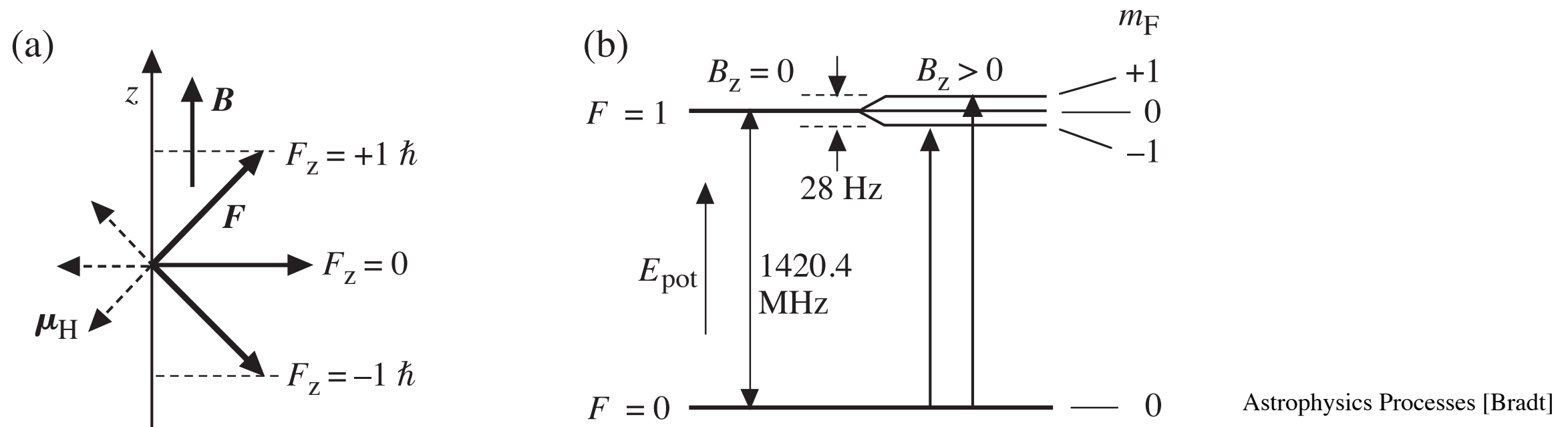
$\mu \simeq \mu_e$ because μ arises almost totally from the electron.

The separation between the +1 and -1 levels is

$$\Delta E_{\text{pot}} = E_{\text{pot}}(m_F = +1) - E_{\text{pot}}(m_F = -1) = g_e \frac{e\hbar}{m_e} B$$

(2) In the $F = 0$ ground state, the total angular momentum is zero. Therefore, the magnetic moment is zero and no Zeeman splitting occurs in the ground state.

Photons carry one unit of angular momentum $\mathbf{J}$. The allowed transitions are from the $F = 0$ ($m_F = 0$) to the $F = 1$ ($m_F = +1$) or $F = 1$ ($m_F = -1$).



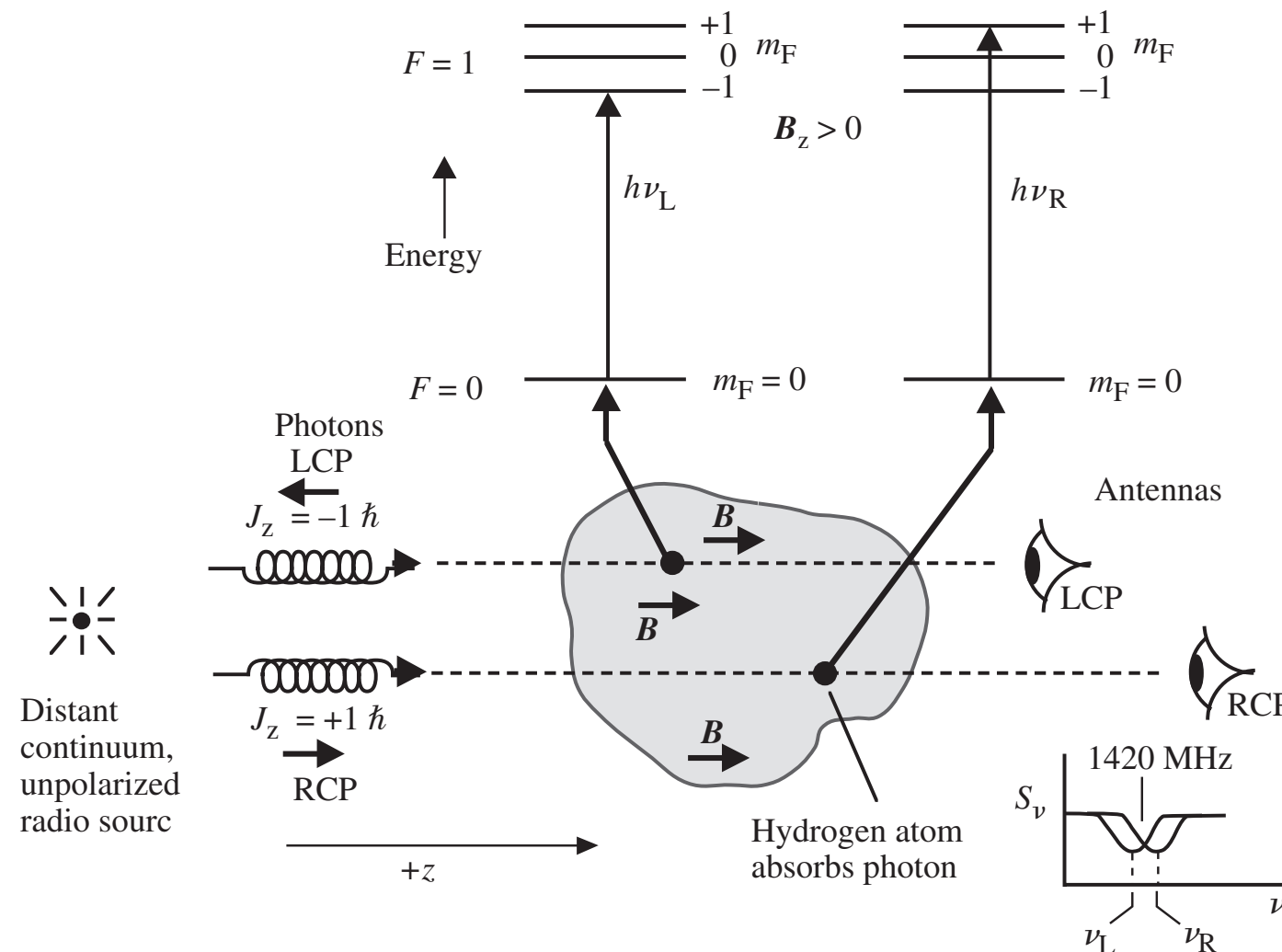
Energy levels for Zeeman absorption by hydrogen atom in an external magnetic field $\mathbf{B}$. The z -axis is taken to be in the direction of $\mathbf{B}$.

- (a) The three allowed projections ($m_F = +1, 0, -1$) of the angular momentum for the $F = 1$ state. The atomic magnetic moment μ_H , which is directed opposite to the angular momentum, interacts with the magnetic field to perturb the energy state.
- (b) Energy levels showing (left) the 1420-MHz transition for $B = 0$ and (right) the Zeeman splitting of the upper energy level into the three sublevels for $B_z > 0$. The two allowed transitions to the upper states are shown; they differ in frequency by 28 Hz for $B_z = 1 \text{ nT} (= 10\mu\text{G})$.

The typical magnetic field strength in the ISM is $\sim 3\text{--}6 \mu\text{G}$ near the Sun, and is higher in dense clouds and near the Galactic center ($\sim 10\text{--}100 \mu\text{G}$).

In quantum theory, a single photon is always circularly polarized with $S = \pm \hbar$ (no state with $S = 0$). Linearly polarized photon state is a superposition of a pair of circularly polarized photons.

A magnetic field of order $10 \mu\text{G}$ shifts the frequency by about one part in 10^8 of the hyperfine splitting. This shift is much smaller than the frequency shift $v/c \sim 10^{-5}$ due to a radial velocity of a few km s^{-1} , and it would be nearly impossible to detect, except that it leads to a shift in frequency between two circular polarization modes. The Zeeman effect in H I 21 cm can therefore be detected by taking the difference of the two circular polarization signals. This technique has been used to measure the magnetic field strength in a number of H I regions.



$$\nu_R - \nu_L = \frac{eB_z}{2\pi m_e} = 28.0 \left(\frac{B_z}{1.0 \text{ nT}} \right) \text{ Hz}$$

Left circularly polarized (LCP) photons are absorbed at a slightly lower frequency.
 Right circularly polarized (RCP) photons are absorbed at a slightly higher frequency.
 (Bradt: Astrophysics Processes)

See also *Heiles & Robishaw (2009)*

Line Broadening

Plasma Effects

Introduction

- So far we have assumed our propagation medium to be a vacuum.
- **Dispersion**: The electrons and magnetic fields of the ionized interstellar medium (ISM) alter the character of radio waves propagating through it. The effect of the electron is to slow the propagation speed of the low-frequency components of a signal relative to that of the higher frequencies. **This phenomenon, the variation of transmission speed with frequency through a plasma, is known as dispersion.**
 - A pulse of radiation emitted by a radio pulsar will contain a wide range of radio frequencies (e.g., 100-400 MHz). During the pulse's passage through the ISM, the higher frequencies propagate faster than the lower ones. A receiver on earth tuned to 100 MHz will detect the pulse substantially delayed relative to the arrival time at 400 MHz.
 - Radio pulsars emit sequential pulses of radiation, and so the dispersion is repeated again and again. One could not observe dispersion with a steady source; all frequencies would arrive at all times.
- Roughly speaking a **plasma** is a globally neutral partially or completely ionized gas.

[Dispersion in Cold, Isotropic Plasma]

- Maxwell's equations for a vacuum can be used for a plasma if the charge and current densities ρ and $\mathbf{j}$ due to the plasma are explicitly included.
- Plasma Dispersion Relation**

If we assume a space and time variation of all quantities of the form $\exp[i(\mathbf{k} \cdot \mathbf{r} - \omega t)]$, Maxwell's equations become

Gauss's law Gauss's law for magnetic field Ampere's law Faraday's law	$\nabla \cdot \mathbf{E} = 4\pi\rho$ $\nabla \cdot \mathbf{B} = 0$ $\nabla \times \mathbf{E} = -\frac{1}{c} \frac{\partial \mathbf{B}}{\partial t}$ $\nabla \times \mathbf{B} = \frac{4\pi}{c} \mathbf{j} + \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t}$	$\longrightarrow$	$i\mathbf{k} \cdot \mathbf{E} = 4\pi\rho$ $i\mathbf{k} \cdot \mathbf{B} = 0$ $i\mathbf{k} \times \mathbf{E} = i\frac{\omega}{c} \mathbf{B}$ $i\mathbf{k} \times \mathbf{B} = \frac{4\pi}{c} \mathbf{j} - i\frac{\omega}{c} \mathbf{E}$
---	--	-------------------	---

Let us assume that the our plasma consists of electrons with density n . The ions are very much less mobile than electrons and thus are neglected. We also assume that there is no external magnetic field; thus the plasma is isotropic. Then, the equation of motion of electrons when there is no external magnetic field is

$$m_e \dot{\mathbf{v}} = -e \left(\mathbf{E} + \frac{\mathbf{v}}{c} \times \mathbf{B} \right) = -e\mathbf{E} \quad \longrightarrow \quad \begin{aligned} -im_e\omega\mathbf{v} &= -e\mathbf{E} \\ \mathbf{v} &= \frac{e\mathbf{E}}{i\omega m_e} \end{aligned}$$

Current density: $\mathbf{j} = -nev = \sigma \mathbf{E}$ where the **conductivity** is $\sigma \equiv \frac{ine^2}{\omega m_e}$.

The charge conservation equation (continuity equation) gives

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0 \qquad -i\omega\rho + i\mathbf{k} \cdot \mathbf{j} = 0 \quad \rightarrow \quad \rho = \frac{1}{\omega} \mathbf{k} \cdot \mathbf{j} = \frac{\sigma}{\omega} \mathbf{k} \cdot \mathbf{E}$$

Using the expressions for $\mathbf{j}$ and ρ , Maxwell's equations become

$$\begin{array}{ccc} \mathbf{j} = \sigma \mathbf{E} & & \\ \rho = \frac{\sigma}{\omega} \mathbf{k} \cdot \mathbf{E} & \longrightarrow & \begin{array}{l} i\mathbf{k} \cdot \mathbf{E} = 4\pi\rho \\ i\mathbf{k} \cdot \mathbf{B} = 0 \\ i\mathbf{k} \times \mathbf{E} = i\frac{\omega}{c} \mathbf{B} \\ i\mathbf{k} \times \mathbf{B} = \frac{4\pi}{c} \mathbf{j} - i\frac{\omega}{c} \mathbf{E} \end{array} \longrightarrow \end{array}$$

These equations are now “source-free” and can be solved in the same way as before. We find again that $\mathbf{k}$, $\mathbf{E}$, $\mathbf{B}$ form a mutually orthogonal right-hand vector triad.

The **dispersion relation** connecting k and ω is given by

$$\begin{array}{l} i\mathbf{k} \times \mathbf{B} = -i\frac{\omega}{c} \epsilon \mathbf{E} \\ i\mathbf{k} \times \left(\frac{c}{\omega} \mathbf{k} \times \mathbf{E} \right) = -i\frac{\omega}{c} \epsilon \mathbf{E} \\ \mathbf{k} (\mathbf{k} \cdot \mathbf{E}) - \mathbf{E} (\mathbf{k} \cdot \mathbf{k}) = -\frac{\omega^2}{c^2} \epsilon \mathbf{E} \end{array} \longrightarrow \begin{array}{l} c^2 k^2 = \epsilon \omega^2 \\ \qquad \qquad = \omega^2 - \omega_p^2 \\ \omega^2 = \omega_p^2 + c^2 k^2 \end{array}$$

$$i\mathbf{k} \cdot \epsilon \mathbf{E} = 0$$

$$i\mathbf{k} \cdot \mathbf{B} = 0$$

$$i\mathbf{k} \times \mathbf{E} = i\frac{\omega}{c} \mathbf{B}$$

$$i\mathbf{k} \times \mathbf{B} = -i\frac{\omega}{c} \epsilon \mathbf{E}$$

where the **dielectric constant**:

$$\begin{aligned} \epsilon &\equiv 1 - \frac{4\pi\sigma}{i\omega} = 1 - \frac{4\pi n e^2}{m_e \omega^2} \\ &= 1 - \left(\frac{\omega_p}{\omega} \right)^2 \end{aligned}$$

plasma frequency:

$$\begin{aligned} \omega_p &\equiv \sqrt{\frac{4\pi n e^2}{m_e}} \\ &= 5.63 \times 10^4 \left(n / \text{cm}^{-3} \right)^{1/2} \text{ s}^{-1} \end{aligned}$$

-
- We now see immediately that for $\omega < \omega_p$, the wavenumber is imaginary ($k^2 < 0$).

$$k = \frac{i}{c} \sqrt{\omega_p^2 - \omega^2} \qquad e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \propto e^{-|k|r}$$

In this case, the wave amplitude decreases as $e^{-|k|r}$ on a scale of the order of $1/|k| \approx c/\omega_p$.

Thus the plasma frequency ω_p defines **a plasma cutoff frequency below which there is no electromagnetic propagation**.

For instance, Earth ionosphere prevents extraterrestrial radiation at frequencies less than ~ 1 MHz from being observed at the earth's surface ($n \sim 10^4 \text{ cm}^{-3}$). The existence of the plasma cutoff yields an important method of probing the ionosphere.

Radio astronomy from the earth's surface can not be carried out at frequencies $\nu \lesssim 9$ MHz. Such radio waves from outer space are reflected back to space by the ionosphere.

Method of probing the ionosphere:

Let a pulse of radiation in a narrow range about ω be directed straight upward from the earth's surface.

When there is a layer at which the density n is large enough to make $\omega < \omega_p$, the pulse will be totally reflected from the layer.

The time delay of the pulse provides information on the height of the layer.

By making such measurements at many different frequencies, the electron density can be determined as a function of height.

• Radio Broadcast

AM radio ($\sim 500\text{--}1700\text{ kHz}$) can propagate long distances via such reflections from the bottom of the ionosphere. This is particularly effective at night when the relatively low “D” layer of the ionosphere becomes neutral (and disappears); then reflection can take place at higher altitudes that have large values of n_e .

In contrast, FM radio at $\sim 100\text{ MHz}$ travels right through the ionosphere into outer space; it sounds better, but one has to be close enough to the station that the signal can be received more or less directly without ionosphere reflections.

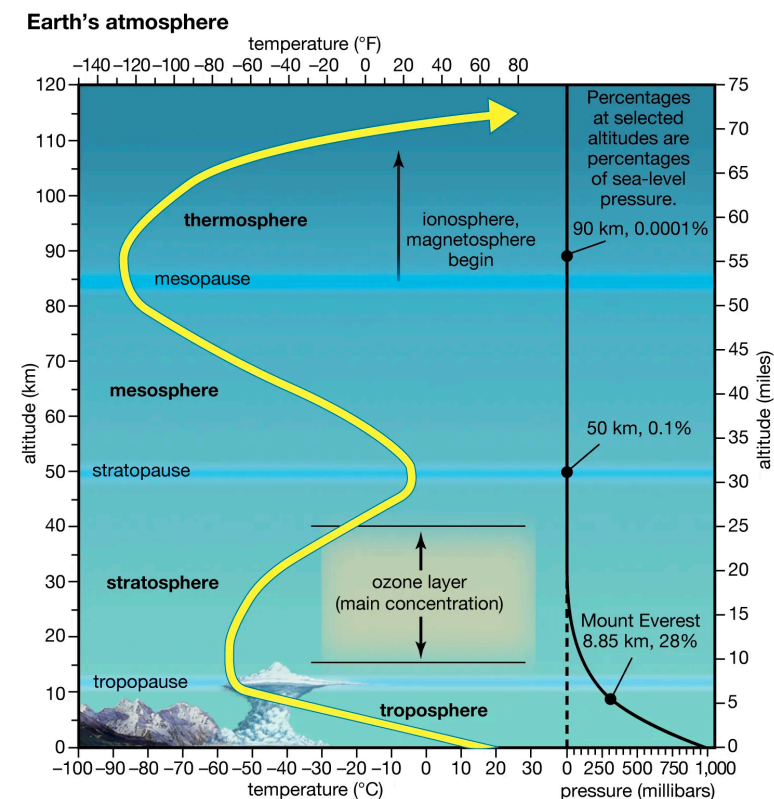
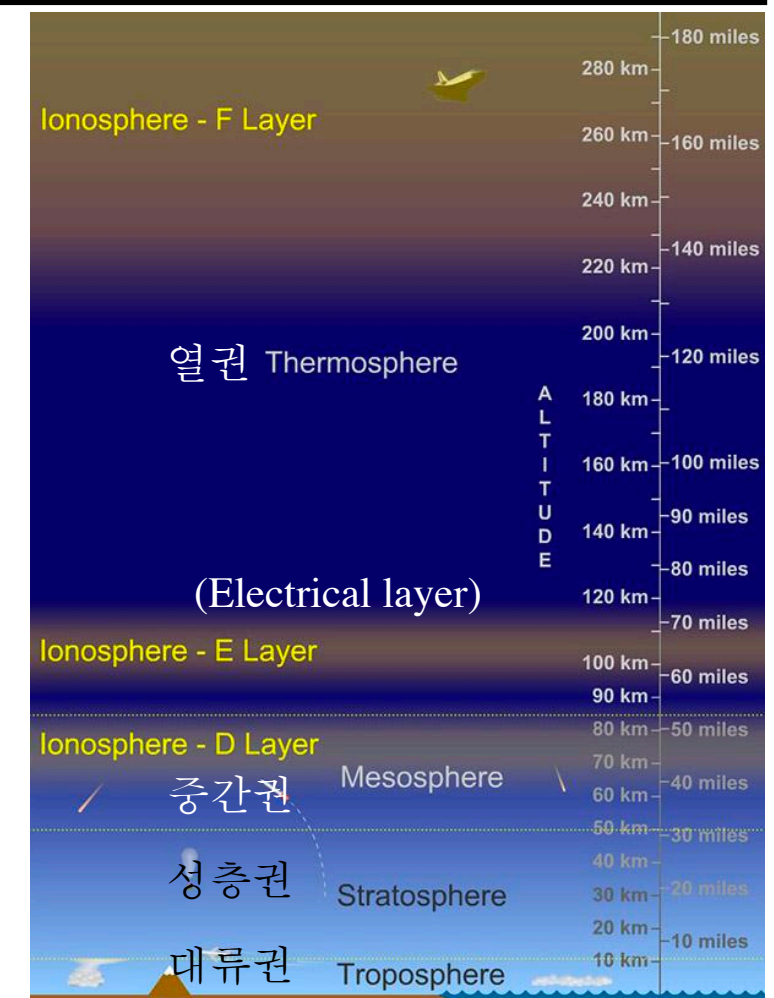
The ionosphere is between 80 and $\sim 600\text{ km}$ where EUV and X-ray solar radiation ionize the atoms and molecules thus creating a layer of electrons.

• Interstellar cutoff

In interstellar space, the number densities of ions are typically much less ($n_e \sim 1\text{ cm}^{-3}$); thus

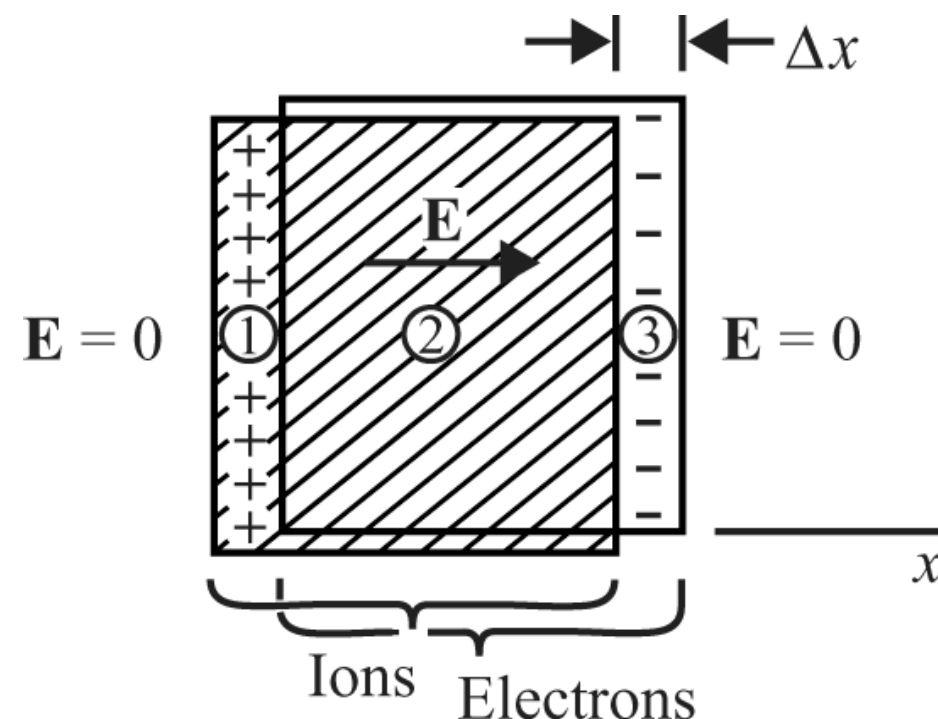
$$\nu_p \approx 300 - 1000\text{ Hz}$$

This is well below the frequencies that penetrate the ionosphere ($\nu_p \gtrsim 9\text{ MHz}$). Consequently, **radio waves that can penetrate the atmosphere can easily travel through much of interstellar space. This makes galactic and extragalactic radio astronomy possible.**



[Plasma Frequency]

- If the electrons in a uniform, homogeneous plasma are displaced from their equilibrium position, an electric field arises because of charge separation. This electric field produces a restoring force on the displaced electrons. Since the magnitude of the charge imbalance is directly proportional to the displacement, the restoring force is given by Hooke's law, $F = -k\Delta x$ and the system behaves as a harmonic oscillator. The resulting oscillations are called **electron plasma oscillations**. The oscillation frequency is called the **electron plasma frequency**.



$$\nabla \cdot \mathbf{E} = 4\pi\rho \quad \rightarrow \quad E = 4\pi ne\Delta x$$

n = electron density

$$m_e \frac{d^2 \Delta x}{dt^2} = -eE = -4\pi ne^2 \Delta x$$

$$\frac{d^2 \Delta x}{dt^2} + \frac{4\pi ne^2}{m_e} \Delta x = 0$$

$$\frac{d^2 \Delta x}{dt^2} + \omega_p^2 \Delta x = 0$$

$$\text{where } \omega_p \equiv \frac{4\pi ne^2}{m_e}$$

[Phase Velocity and Group Velocity]

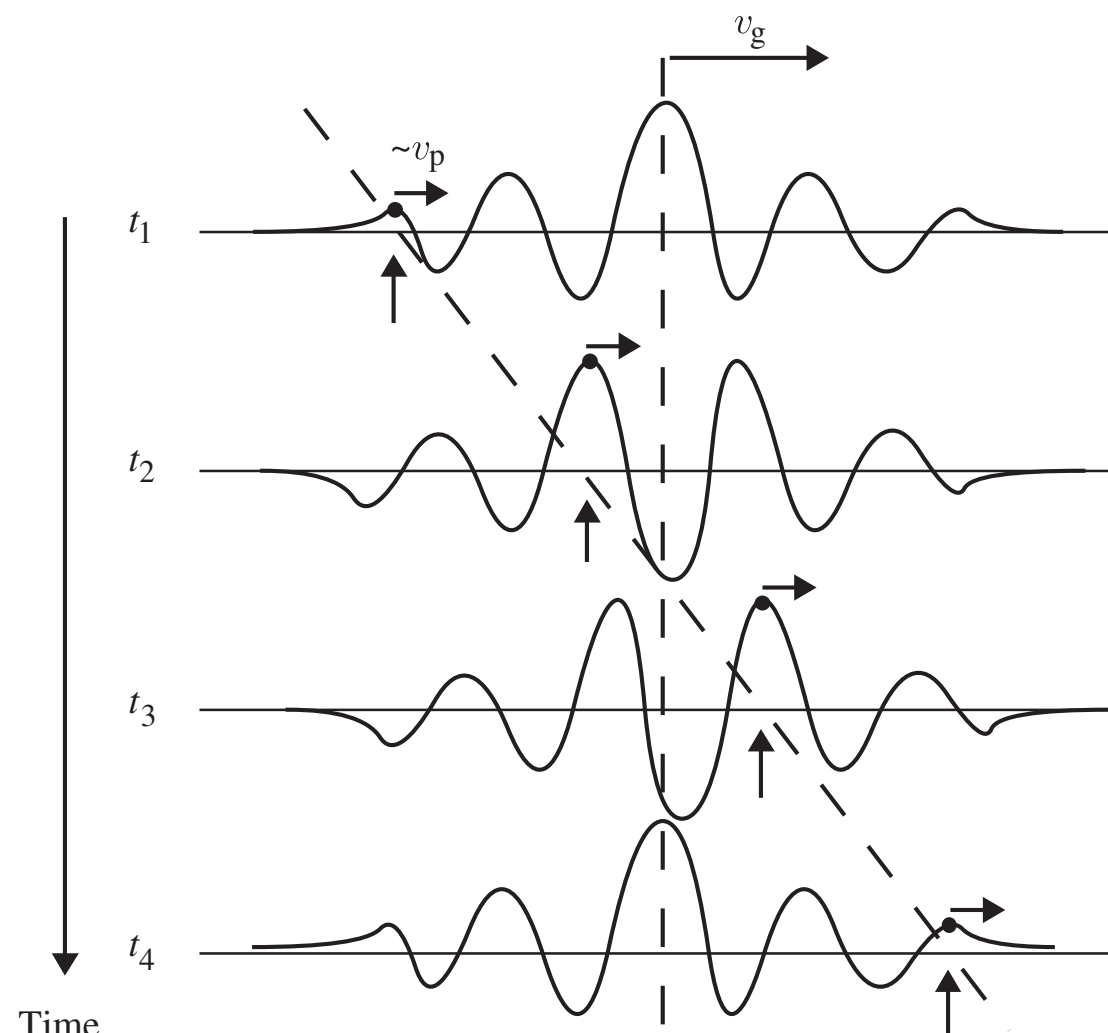
- Dispersion relation, phase velocity, and group velocity:

dispersion relation = a function which gives ω as a function of k .

phase velocity = the rate at which the phase of the wave propagates in space.

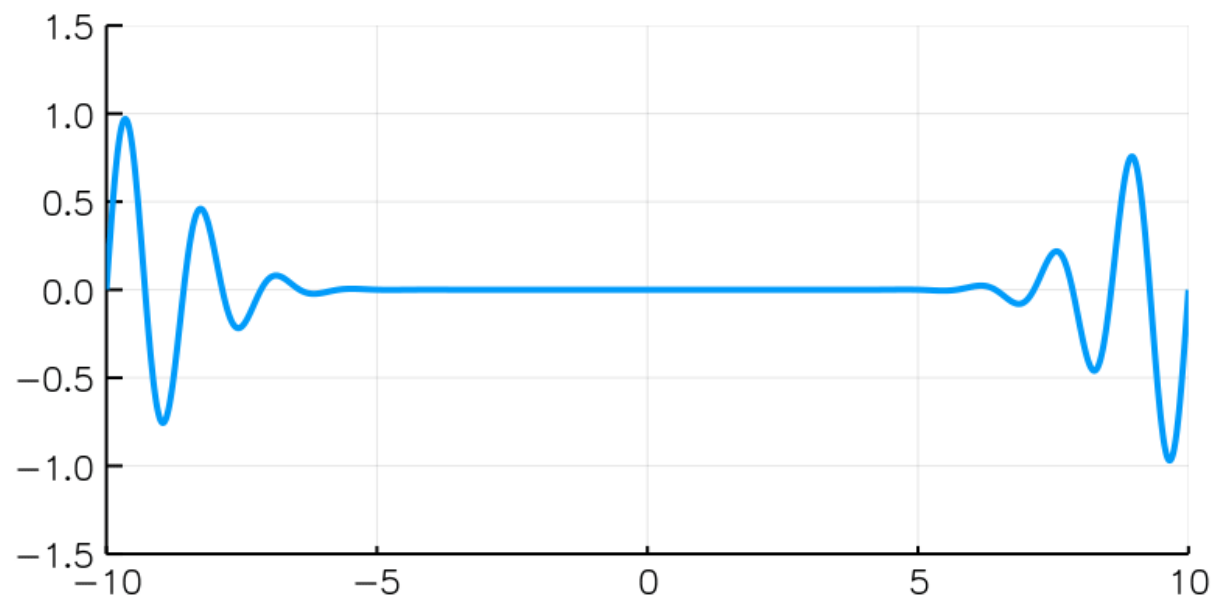
group velocity = the velocity with which the overall shape of the waves' amplitudes (modulation or envelope of the wave) propagates through space.

$$\text{In vacuum, } \omega = ck, \quad v_{\text{ph}} \equiv \frac{\omega}{k} = c, \quad v_g \equiv \frac{\partial \omega}{\partial k} = c$$



A wavelet or wave packet shown at four different times. The packet as a whole moves to the right at group velocity v_g . The interference between the component waves that make up the packet, with their different phase velocities, causes a single crest (arrows and dots) to gradually overtake the centroid of the packet. Relative to the medium, the overtaking crest moves at approximately the median phase velocity.

Figure 11.3. [Bradt] Astrophysics Processes



https://en.wikipedia.org/wiki/Group_velocity

- Assume that a wave packet E is almost monochromatic, so that its Fourier component is nonzero only in the vicinity of a central wavenumber k_0 . Then, linearization gives:

$$\begin{aligned}\omega(k) &\approx \omega_0 + (k - k_0) \left. \frac{\partial \omega}{\partial k} \right|_{k=k_0} \\ &= \omega_0 + (k - k_0)v_g\end{aligned}$$



$$\begin{aligned}E(x, t) &= \int dk \int d\omega \bar{E}(k, \omega) e^{i(kx - \omega t)} \\ &\approx e^{it(v_g k_0 - \omega_0)} \int dk \bar{E}(k, \omega_0) e^{ik(x - v_g t)}\end{aligned}$$

where $\boxed{v_g \equiv \left. \frac{\partial \omega}{\partial k} \right|_{k=k_0}}$

$$\boxed{|E(x, t)| = |E(x - v_g t, 0)|}$$

- Therefore, the envelope of the wavepacket travels at velocity $v_g = \left(\partial \omega / \partial k \right)_{k=k_0}$.

- **Group and Phase Velocity in terms of the index of refraction**

Phase velocity

$$v_{\text{ph}} \equiv \frac{\omega}{k} = \frac{\omega}{\sqrt{\omega^2 - \omega_p^2}/c} = \frac{c}{n_r}$$

where *the index of refraction* is defined to be

$$n_r \equiv \sqrt{\epsilon} = \sqrt{1 - \omega_p^2/\omega^2}$$

When $\omega > \omega_p$, waves do propagate without damping with phase velocity and the phase velocity always exceeds the speed of light ($v_{\text{ph}} > c$).

Group velocity

Group velocity is the speed with which a pulse of radiation containing a band of frequencies will propagate through space.

$$\omega^2 = \omega_p^2 + c^2 k^2 \quad \longrightarrow \quad v_g \equiv \frac{\partial \omega}{\partial k} = \frac{c^2 k}{\omega} = c \sqrt{1 - \frac{\omega_p^2}{\omega^2}} \quad \rightarrow \quad v_g < c$$

$$\begin{aligned} v_{\text{ph}} &= \frac{c}{n_r} \\ v_g &= c n_r \end{aligned}$$

The group velocity is always less than c . The wave energy travels at the group velocity, as does any modulation of the wave (information coding). **The information content moves at the group velocity.**

The lower frequencies travel more slowly than higher frequencies. For $\omega < \omega_p$, the index of refraction becomes imaginary, leading to attenuation. If the frequency ω is lowered toward ω_p and increasing, the group velocity decreases toward zero. If the frequency ω is well above ω_p and increasing, the group velocity approaches c . As the density is lowered ($n \rightarrow 0$; become more closely a vacuum), ω_p decreases and the group velocity more closely approaches the vacuum velocity c .

[Dispersion Measure]

- **Pulsar**

An important application of the formula for group velocity is to pulsars. Each individual pulse from the pulsar has a spectrum covering a wide band of frequency. Thus, the pulse will be dispersed by its interaction with the interstellar plasma, since each small range of frequencies travels at a slightly different group velocity and will reach earth at a slightly different time.

Let d = pulsar distance to the pulsar

The time required for a pulse to reach earth at frequency ω is $t_p = \int_0^d \frac{ds}{v_g}$.

The plasma frequency in ISM is usually quite low ($\sim 10^3$ Hz), so we can assume that $\omega \gg \omega_p$. Then, the travel time is given by

$$\frac{1}{v_g} = \frac{1}{c} \left(1 - \frac{\omega_p^2}{\omega^2} \right)^{-1/2} \approx \frac{1}{c} \left(1 + \frac{1}{2} \frac{\omega_p^2}{\omega^2} \right) \longrightarrow \boxed{t_p \approx \frac{d}{c} + \frac{2\pi e^2}{cm_e \omega^2} \int_0^d n ds}$$

= transit time for a vacuum + plasma correction

- **Dispersion measure:**

What is usually measured is **the rate of change of arrival time with respect to frequency**.

$$\boxed{\frac{dt_p}{d\omega} = -\frac{4\pi e^2}{cm_e \omega^3} \mathcal{DM} \text{ where } \mathcal{DM} \equiv \int_0^d n ds}$$

is the dispersion measure of the ray.

This is the electron column density.

The arrival interval of pulsar signal between two frequencies ν_1 and ν_2 :

$$\boxed{\Delta t_p = \frac{e^2}{2\pi cm_e} \left(\frac{1}{\nu_1^2} - \frac{1}{\nu_2^2} \right) \mathcal{DM} = 4.15 \times 10^{15} \left(\frac{1}{\nu_1^2} - \frac{1}{\nu_2^2} \right) \frac{\mathcal{DM}}{\text{cm}^{-3} \text{ pc}} \text{ sec}}$$

- **Crab nebula**

Observations of the pulses from the Crab pulsar show that the delay between frequencies 100 MHz and 400 MHz is 23.5 s, which yields a value for the column density expressed as

$$\int_0^D n_e ds = 1.86 \times 10^{20} \text{ cm}^{-2}. \quad \text{electron column density to Crab pulsar}$$

Note that astronomers use the mixed units pc cm⁻³ to denote the dispersion measure (DM), but cm⁻² for the column density.

Because the Crab pulsar is about 2 kpc ($= 6.2 \times 10^{21}$ cm), the average electron density is

$$\langle n_e \rangle_{\text{abs}} = \frac{1}{D} \int_0^D n_e ds = 0.03 \text{ electrons cm}^{-3}$$

This is relatively low density in the context of the ISM. It lies between the densities of the Warm Ionized Medium (WIM) and Hot Ionized Medium (HIM).

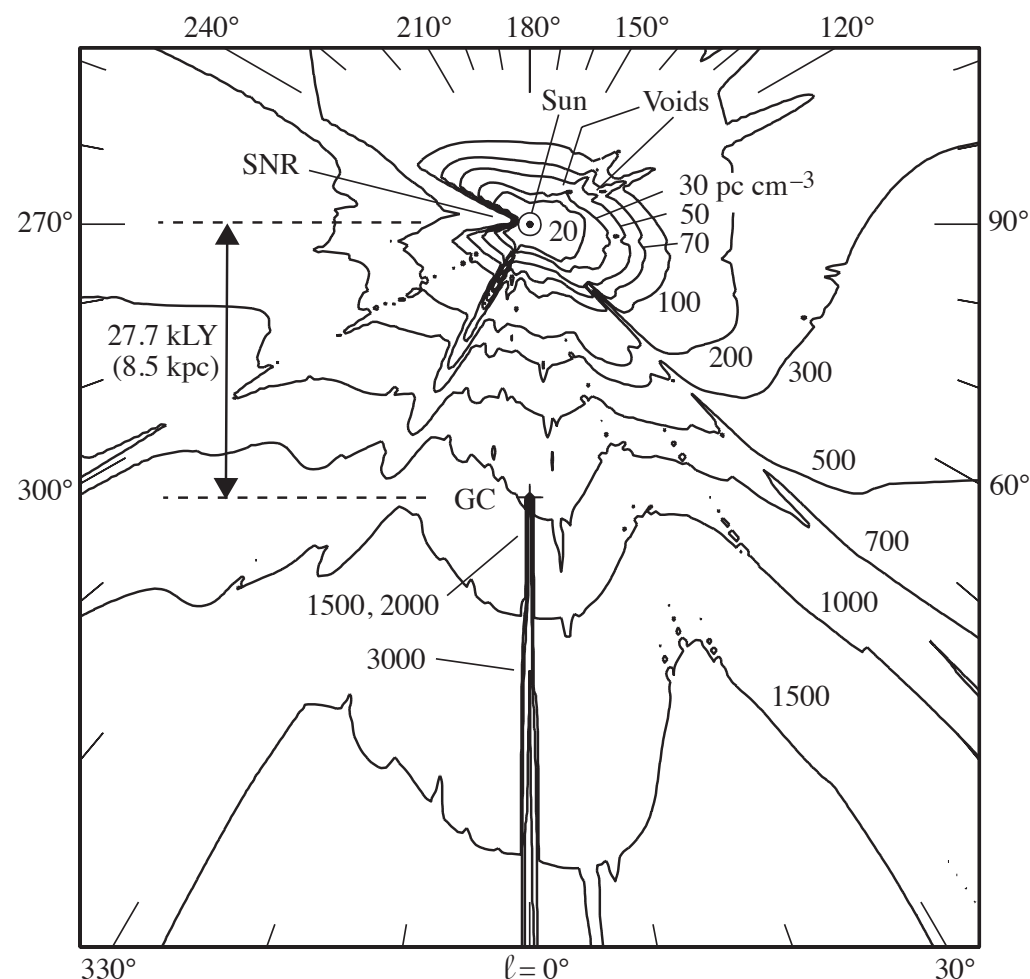
The 23-s time lag from 400 MHz to 100 MHz is very long compared with the 33-ms interval between pulses from the Crab pulsar. At earth, each pulse is “heard” as a frequency that continuously decreases.

- **Galactic model of electron density**

The number of known radio pulsars is now over 2000, and the DM has been measured for almost all of them. In some cases, the distance to the pulsar is known from independent evidence such as parallax measurements, H I absorption, or because the distance is associated with an object of known distance such as a supernova remnant. The DM in these cases provides an average value of the electron density along the path.

$$\mathcal{DM} = \int n_e ds \quad \mathcal{EM} = \int n_e^2 ds$$

The DM and other results (such as H α) have led to a model of electron density n_e in the Galaxy. The best current model for the electron density n_e for the Galactic plane is known as NE2001 because it is constructed from data available through the year 2001.



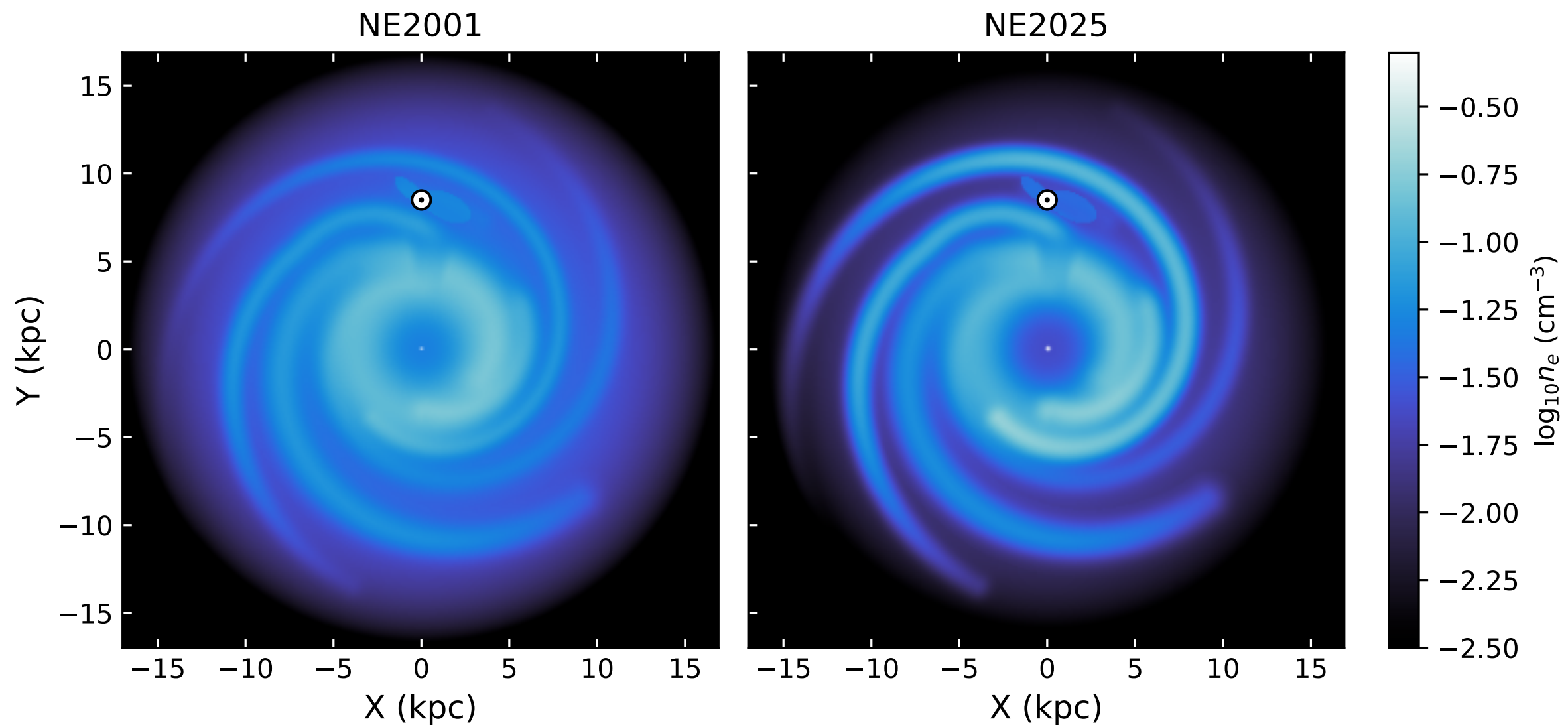
Contours of DM in units of pc cm⁻³ in the Galactic plane viewed from above the plane.

The sun and Galactic center (GC, cross) are marked; their separation is 8.5 pc.

Strong local enhancements of n_e , such as the GC at $\ell = 0$ or the Gum and Vela nebulae (SNR) at $\ell \approx 260^\circ$, are reflected in the contours at distances beyond the enhancements. Similarly, voids such as those $\ell \approx 120^\circ$ produce “fingers” in the contours at greater distance

Ocker & Cordes 2025, arXiv:2602.11838

NE2025 - <https://github.com/stella-ocker/mwprop>



Electron density in the Galactic plane ($z = 0$) for NE2001 (left) and NE2025 (right). The color stretch is the same for both panels.

Figure 1 in Ocker & Cordes (2026)

[References]

Taylor & Cordes (1993, ApJ, 411, 674)

Cordes & Lazio (2003, arXiv:astro-ph/0207156)

Schnitzeler (2012, MNRAS, 427, 664)

Ocker & Cordes (2026, ApJ, arXiv:2602.11838)

[Faraday Rotation - Propagation along a Magnetic Field]

- Now we extend the discussion of plasma propagation effects by considering the effect of an external, fixed magnetic field $\mathbf{B}_0$.

The properties of the waves will then depend on the direction of propagation relative to the magnetic field direction. For this reason the plasma is called *anisotropic*.

Because of the magnetic field, a new frequency enters the problem, namely, the cyclotron frequency.

$$\begin{aligned} \text{cyclotron frequency} \quad \omega_B &= \frac{eB_0}{m_e c} = 1.67 \times 10^7 (B_0/\text{G}) \text{ s}^{-1} \\ \hbar\omega_B &= 1.16 \times 10^{-8} (B_0/\text{G}) \text{ eV} \end{aligned}$$

If the fixed magnetic field $\mathbf{B}_0$ is much stronger than the field strengths of the propagating wave, then the equation of motion of an electron is approximately

$$m_e \frac{d\mathbf{v}}{dt} = -e\mathbf{E} - \frac{e}{c} \mathbf{v} \times \mathbf{B}_0$$

For simplicity, assume that **the wave propagates along the fixed field** $\mathbf{B}_0 = B_0 \hat{\mathbf{z}}$, and assume that **the wave is circularly polarized and sinusoidal**.

$$\mathbf{E}_{\pm}(t) = E e^{-i\omega t} (\hat{\mathbf{x}} \pm i\hat{\mathbf{y}})$$

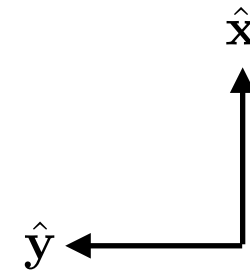
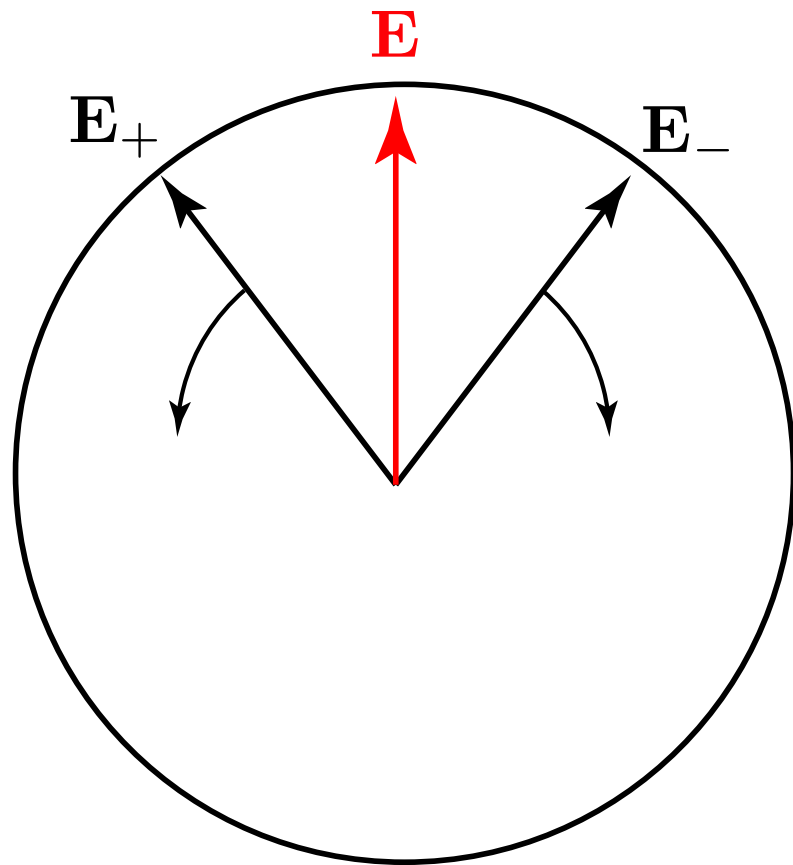
Here, $\pm$ denotes right and left circular polarization.

Note that Rybiki & Lightman define oppositely the RCP and LCP.

Bases vectors for circularly rotating electric field

The bases vectors $\mathbf{E}_{\pm}$ rotates counterclockwise (+) or clockwise (-) when viewed from the tip of the wave vector.

$$\begin{aligned}
 \text{Re}(\mathbf{E}_{\pm}) &= \text{Re}((\hat{\mathbf{x}} \pm i\hat{\mathbf{y}}) E e^{-i\omega t}) \\
 &= \text{Re}((\hat{\mathbf{x}} \pm i\hat{\mathbf{y}}) E (\cos \omega t - i \sin \omega t)) \\
 &\propto (\cos \omega t) \hat{\mathbf{x}} \pm (\sin \omega t) \hat{\mathbf{y}} \\
 &\propto \cos(\pm \omega t) \hat{\mathbf{x}} + \sin(\pm \omega t) \hat{\mathbf{y}}
 \end{aligned}$$



Now, we solve the equation of motion for an electron and obtain the dielectric constant.

$$m_e \frac{d\mathbf{v}}{dt} = -e\mathbf{E} - \frac{e}{c}\mathbf{v} \times \mathbf{B}_0$$

$$\boxed{\mathbf{E} \perp \mathbf{B}_0 \longrightarrow \frac{d\mathbf{v}_{\parallel}}{dt} = 0 \quad \mathbf{v}_{\parallel} = \text{constant} \\ \mathbf{v}_{\perp} = v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}}}$$

$$(-i\omega)e^{-i\omega t}(v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}}) = -\frac{e}{m_e}Ee^{-i\omega t}(\hat{\mathbf{x}} \pm i\hat{\mathbf{y}}) - \frac{e}{m_e c}e^{-i\omega t}(v_y B_0 \hat{\mathbf{x}} - v_x B_0 \hat{\mathbf{y}})$$

$$\longrightarrow v_x = -\frac{ie}{m_e \omega}E - \frac{ieB_0}{m_e c \omega}v_y, \quad v_y = \pm \frac{e}{m_e \omega}E + \frac{ieB_0}{m_e c \omega}v_x$$

$$\begin{aligned} (1) \quad v_x &= -\frac{ie}{m_e \omega}E - \frac{ieB_0}{m_e c \omega} \left(\pm \frac{e}{m_e \omega}E + \frac{ieB_0}{m_e c \omega}v_x \right) & (2) \quad v_y &= \pm \frac{e}{m_e \omega}E + \frac{ieB_0}{m_e c \omega} \left(-\frac{ie}{m_e \omega}E - \frac{ieB_0}{m_e c \omega}v_y \right) \\ \rightarrow \left(1 - \frac{e^2 B_0^2}{m_e^2 c^2 \omega^2} \right) v_x &= -\frac{ie}{m_e \omega}E \left(1 \pm \frac{eB_0}{m_e c \omega} \right) & \rightarrow \left(1 - \frac{e^2 B_0^2}{m_e^2 c^2 \omega^2} \right) v_y &= \pm \frac{e}{m_e \omega}E \left(1 \pm \frac{eB_0}{m_e c \omega} \right) \\ \rightarrow (\omega^2 - \omega_B^2) v_x &= -\frac{ie}{m_e}E (\omega \pm \omega_B) & \rightarrow (\omega^2 - \omega_B^2) v_y &= \pm \frac{e}{m_e}E (\omega \pm \omega_B) \\ \rightarrow v_x &= -\frac{ie}{m_e}E \frac{1}{\omega \mp \omega_B} & \rightarrow v_y &= \pm \frac{e}{m_e}E \frac{1}{\omega \mp \omega_B} \end{aligned}$$

Finally, the current density and conductivity are obtained to be

$$\therefore \mathbf{v} = \frac{-ie}{m_e (\omega \mp \omega_B)} \mathbf{E}(t)$$

$$\mathbf{j} \equiv -ne\mathbf{v} = \sigma \mathbf{E}$$

$$\text{where } \sigma_{R,L} \equiv \frac{ine^2}{m_e (\omega \mp \omega_B)}$$

Dielectric constant

$$\epsilon \equiv 1 - \frac{4\pi\sigma}{i\omega} = 1 - \frac{4\pi ne^2}{m_e\omega(\omega \mp \omega_B)}$$

$$\epsilon_{R,L} = 1 - \frac{\omega_p^2}{\omega(\omega \mp \omega_B)}$$

Dispersion relation

$$\omega^2 = k^2 c^2 + \frac{\omega_p^2 \omega}{\omega \mp \omega_B}$$

Phase velocity

$$v_p \equiv \frac{\omega}{k} = \frac{c}{n_r}$$

$$n_r = \sqrt{1 - \frac{\omega_p^2}{\omega(\omega \mp \omega_B)}}$$

Right (+) and left (-) circularly polarized waves travel with different speeds.

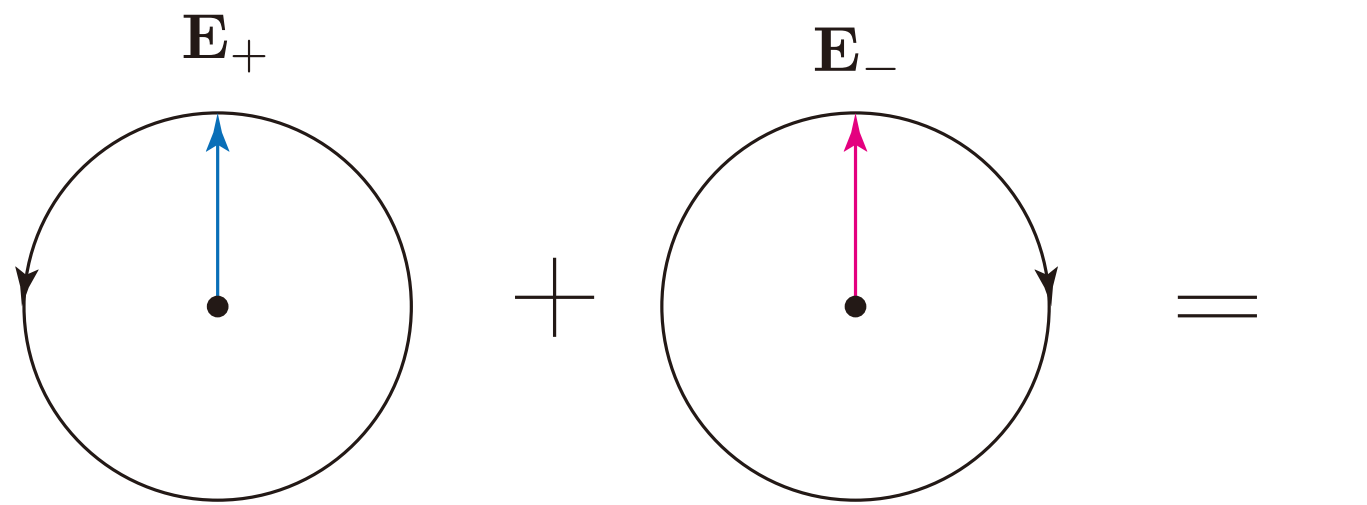
Speed difference sense is $v_R > v_L$.

These waves travel with different velocities. Therefore, a plane polarized wave, which is a linear superposition of a right-hand and a left-hand polarized wave, will not keep a constant plane of polarization, but **this plane will rotate as it propagates**. This effect is called **Faraday rotation**.

- **Faraday Rotation**

If the incident radiation is *circularly polarized* (either R or L), then the radiation will encounter different dispersion than unmagnetized case. But, the radiation will still remain circularly polarized.

If the incident radiation is *linearly polarized*, i.e., a linear superposition of a right-hand and a left-hand polarized wave, then *the line of polarization will rotate as it propagates*. This effect is called Faraday rotation.



$$\begin{aligned}
 \mathbf{E}(t) &= E e^{-i\omega t} \hat{\mathbf{x}} \\
 &= \frac{1}{2} [(\hat{\mathbf{x}} + i\hat{\mathbf{y}}) + (\hat{\mathbf{x}} - i\hat{\mathbf{y}})] E e^{-i\omega t} \\
 &= \frac{1}{2} (\epsilon_+ + \epsilon_-) E e^{-i\omega t}
 \end{aligned}$$

Decomposition of linear polarization into components of right and left circular polarization

- The phase angle ϕ after traveling a distance d is, in general, if the wave number is not constant along the path, is given by

$$\phi_{R,L} = \int_0^d k_{R,L} ds$$

Assume that $\omega \gg \omega_p$ and $\omega \gg \omega_B$

$$\begin{aligned} k_{R,L} &= \frac{\omega}{c} \sqrt{\epsilon_{R,L}} = \frac{\omega}{c} \left[1 - \frac{\omega_p^2}{\omega^2 (1 \mp \omega_B/\omega)} \right]^{1/2} \approx \frac{\omega}{c} \left[1 - \frac{\omega_p^2}{2\omega^2} \left(1 \pm \frac{\omega_B}{\omega} \right) \right] \\ &= \frac{\omega}{c} - \frac{\omega_p^2}{2c\omega} \mp \frac{\omega_p^2 \omega_B}{2c\omega^2} = k_0 \mp \frac{1}{2} \Delta k \quad \left(\Delta k \equiv k_L - k_R = \frac{\omega_p^2 \omega_B}{c\omega^2} \right) \end{aligned}$$

Consider an electromagnetic wave that starts off linearly polarized in the x -direction at the source.

$$\mathbf{E}(t) = E e^{-i\omega t} \hat{\mathbf{x}} = \frac{1}{2} [(\hat{\mathbf{x}} + i\hat{\mathbf{y}}) + (\hat{\mathbf{x}} - i\hat{\mathbf{y}})] E e^{-i\omega t}$$

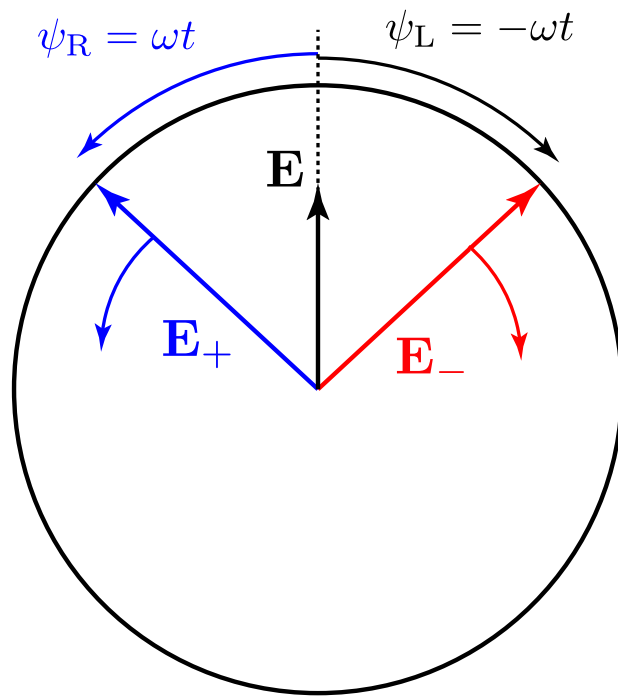
Let's define

$$\int_0^d k_{R,L} ds = \int_0^d k_0 ds \mp \frac{1}{2} \int_0^d \Delta k ds \equiv \phi \mp \varphi$$

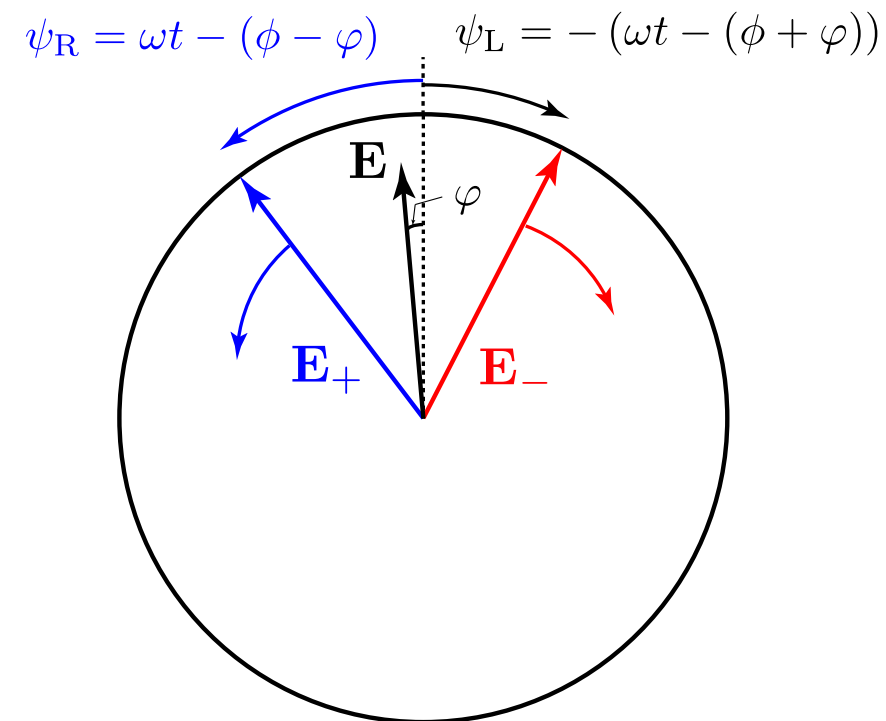
Then, after propagating a distance d through a magnetized plasma toward the observer, the electric field will be

$$\begin{aligned} \mathbf{E}(t) &= \frac{1}{2} \left[(\hat{\mathbf{x}} + i\hat{\mathbf{y}}) \overset{e^{i \int k_R ds}}{e^{i(\phi-\varphi)}} + (\hat{\mathbf{x}} - i\hat{\mathbf{y}}) \overset{e^{i \int k_L ds}}{e^{i(\phi+\varphi)}} \right] E e^{-i\omega t} \\ &= (\hat{\mathbf{x}} \cos \varphi + \hat{\mathbf{y}} \sin \varphi) E e^{i(\phi-\omega t)} \end{aligned}$$

- The resulting electric field is also linearly polarized, but the polarization angle is rotated counterclockwise by angle φ (when viewed at a fixed position).



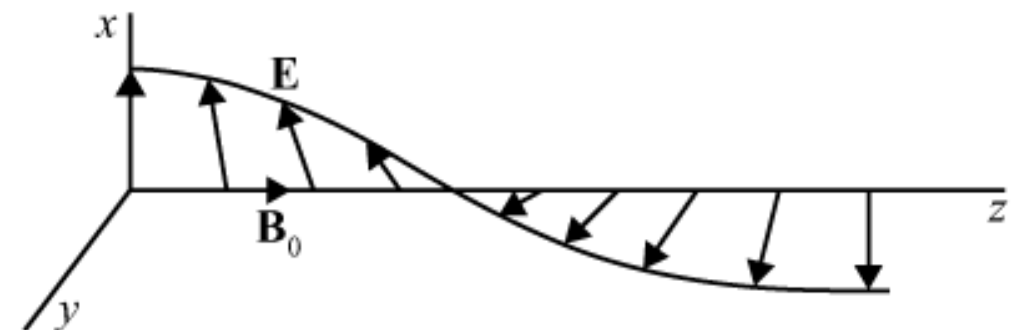
Before passing through the medium



After passing through the medium

- Radiation that starts linearly polarized in a certain direction is rotated by the *Faraday effect* through an angle φ after propagating a distance d through a magnetized plasma.

$$\varphi = \frac{1}{2} \int_0^d \Delta k ds = \frac{1}{2} \int_0^d \frac{\omega_p^2 \omega_B}{c \omega^2} ds = \frac{2\pi e^3}{m_e^2 c^2 \omega^2} \int_0^d n B_{\parallel} ds$$



- We cannot, of course, generally measure the absolute rotation angle, since we do not know the intrinsic polarization direction of the radiation when it started from the source.

However, since φ varies with frequency (as ω^{-2}), we can determine the value of integral $\int nB_{\parallel} ds$ by making measurements at several frequencies. This can give information about the interstellar magnetic field.

Rotation measure is defined by
$$\varphi = \frac{2\pi e^3}{m_e^2 c^2 \omega^2} \mathcal{RM} = \frac{e^3 \lambda^2}{2\pi m_e^2 c^4} \mathcal{RM}, \text{ where } \mathcal{RM} \equiv \int_0^d n_e B_{\parallel} ds$$

However, *the field changes direction often along the line of sight and this method gives only a lower limit to actual field magnitudes.*

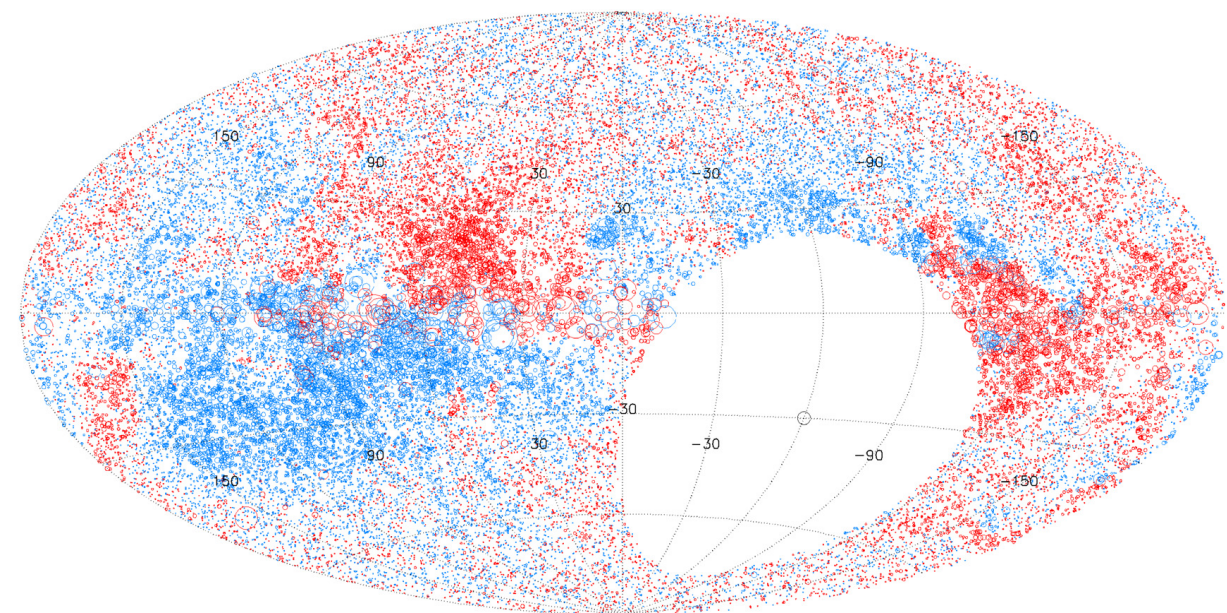
For measurements toward sources (pulsars) where the dispersion measure (DM) is also known, we can derive an estimate of the mean field strength along the line of sight.

$$\langle B_{\parallel} \rangle = \frac{\mathcal{RM}}{\mathcal{DM}} \quad \text{density-weighted magnetic field}$$

Radio astronomers have concluded that

$$\langle n_e \rangle \approx 0.03 \text{ cm}^{-3}$$

$$\langle B_{\parallel} \rangle \approx 3 \mu\text{G}$$



(Taylor, Stil, & Sunstrum 2009, ApJ, 702, 1230)

Red circles are positive RM and blue circles are negative. The size of the circle scales linearly with magnitude of RM.